COURSE TITLE	**BUSINESS MATHEMATICS**
COURSE CODE	**04BC1102**
COURSE CREDITS	**4**

Objective:

1 NA

Course Outcomes: After completion of this course, student will be able to:

1 Understand dimensions of Business Mathematics
2 Formulate different functions and apply them in business problems
3 Integrate and Apply knowledge on Permutation and Combination
4 Formulate different applications of sequence and series and apply them in business problems.
5 Designing the framework of Matrix and Determinants and its usefulness for solving business problems.
6 Analyse how matrices are used as mathematical tools in representing a system of equations.

Pre-requisite of course:NA

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	1	0	50	30	20	0	0

Contents : Unit	Topics	Contact Hours
1	**Introduction to Business Mathematics:** : Scope and Importance, Steps in Quantitative Analysis Approach of Problem Solution, , Number System Basics of Algebra: Algebraic Identity, Basics of Algebra: Algebraic Identity, Equalities and Inequalities, Factorization, Linear and Quadratic Equations, Solution Of Simultaneous Linear Equations, Solution (roots) of quadratic equations, Laws of Indices (Without Proof)	13
2	**Functions and Their Applications** Definition, Types of Functions, Demand Function, Cost Function, Revenue Function, Profit Function, Break Even Point, Applications to Business Problems. Use of MS Excel to Plot Straight Line	10
3	**Permutation and Combination:** Fundamental Rule of counting, Factorial, Permutation, Restricted Permutation, Circular Permutation, Combination, Restricted Combination, Division into Groups. Us, of MS Use of MS Excel to Calculate Permutation and Combination	13

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Contents : Unit	Topics	Contact Hours
4	**Sequence and Series:** Different Types of Sequence, Arithmetic and Geometric Sequence, Arithmetic Progression (AP), Geometric Progression (GP), Arithmetic and Geometric means, Harmonic Progression (HP).	12
5	**Determinant, Matrices and Applications** Introduction, Definition, Types of Matrices, Operations on Matrices, Conversion of Business Problems into a Linear System of Equations (LSE) (Max 3X3) , Determinant of a Matrix, Properties of Determinant, Matrix Equation, , Solution of System of Linear Equations using inverse of coefficient matrix (Max 3), Use of MS Excel to Calculate Determinant and Inverse of Matrix	12
Total Hours		**60**

Textbook :

1 Business Mathematics, P. Mariappan, Pearson Education, 2012

2 A Textbook of Business Mathematics , P. Hazarika, S. Chand Publication, 2014

3 Business Mathematics, D C Sancheti and V K Kapoor, Sultan Chand and Sons, 2010

References:

1 Business Mathematics, Business Mathematics, A. Dikshit and J. Jain, Himalaya Publishing House, 2014

2 Business Mathematics, Business Mathematics, Zamarudeen and Qazi, Vikas Publishing, 2015

3 Business Mathematics, Business Mathematics, Trivedi Kashyap, Pearson Education, 2016

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery and evaluation					
Remember / Knowledge	**Understand**	**Apply**	**Analyze**	**Evaluate**	**Higher order Thinking / Creative**
20.00	30.00	25.00	15.00	10.00	0.00

Instructional Method:

1 NA

Supplementary Resources:

1 NA

2 https://www.coursera.org/learn/mathematical-thinking

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Supplementary Resources:

3 https://www.excel-easy.com/functions/math-functions.html

4 https://www.youtube.com/watch?v=A18v-vxL3TY

5 https://www.youtube.com/watch?v=KoOofjq_fGQ

6 https://www.youtube.com/watch?v=cypZDjH1ImA

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